

Entry: 1

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

When changing the order of the length to

`./scheduler.py -p SJF -l 300,200,100`, the scheduler will execute the last jobs first because it has the shortest length, then the first, then the last.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

7.0

Score 2:

7.0

Score 3:

7.0

Entry: 2

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF: Job 0 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 1 -- Response: 100.00 Turnaround 300.00 Wait 100.00 Job 2 -- Response: 300.00 Turnaround 600.00 Wait 300.00

FIFO: Job 0 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 1 -- Response: 100.00 Turnaround 300.00 Wait 100.00

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19

Score outlier:

8

Entry: 3

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF: Job 0, response 0, turnaround 100 Job 1, response 100, turnaround 300 Job 2, response 300, turnaround 600 FIFO: Job 0, response 0, turnaround 300 Job 1, response 300, turnaround 600 Job 2, response 600, turnaround 900

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

13.0

Score 2:

13.0

Score 3:

19

Entry: 4

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF: Final statistics of 100,200,300: Job 0 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 1 -- Response: 100.00 Turnaround 200.00 Wait 0.00 Job 2 -- Response: 200.00 Turnaround 300.00 Wait 0.00

FIFO: Final statistics of 100,200,300: Job 0 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 1 -- Response: 100.00 Turnaround 200.00 Wait 0.00 Job 2 -- Response: 200.00 Turnaround 300.00 Wait 0.00

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (6.5 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

13.0

Score 2:

13.0

Score 3:

13.0

Score outlier:

5

Entry: 5

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

Both FIFO and SJF response time and turnaround time will be the same if the order is preserved. Response time job 0: 0, job 1: 300, job 2: 500 Turnaround time job 0: 300, job 1: 500, job 2: 600 Average response time job 0: 0, job 1: 300, job 2: 500

FIFO: Response time job 0: 0, job 1: 300, job 2: 500 Turnaround time job 0: 300, job 1: 500, job 2: 600 Average response time job 0: 0, job 1: 300, job 2: 500

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

11

Score 2:

15

Score 3:

13

Entry: 6

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

When the job lengths are 100, 200, 300, the response time and turnaround time of SJF and FIFO are same with each other.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same."
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19.0

Score 2:

19.0

Score 3:

19

Score outlier:

4

Entry: 7

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For FIFO , when I change the order from (100,200,300) to (300, 200, 100), the response time and turn around time will be the same.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same."
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

8

Score 2:

4

Score 3:

6

Entry: 8

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are (./scheduler.py -p SJF -l 100,200,300)

Answer:

It would be different if you change the order of the job length. (e.g. in SJF, it'll prioritise shorter burst time first)

SJF (100,200,300): Execution trace: [time 0] Run job 0 for 100.00 secs (DONE at 100.00) [time 100] Run job 1 for 200.00 secs (DONE at 300.00) [time 300] Run job 2 for 300.00 secs (DONE at 600.00)

SJF (200,100,300): Execution trace: [time 0] Run job 1 for 100.00 secs (DONE at 100.00) [time 100] Run job 0 for 200.00 secs (DONE at 300.00) [time 300] Run job 2 for 300.00 secs (DONE at 600.00)

FIFO (100,200,300): Execution trace: [time 0] Run job 0 for 100.00 secs (DONE at 100.00) [time 100] Run job 1 for 200.00 secs (DONE at 300.00) [time 300] Run job 2 for 300.00 secs (DONE at 600.00)

FIFO (200,100,300): Execution trace: [time 0] Run job 0 for 200.00 secs (DONE at 200.00) [time 200] Run job 1 for 100.00 secs (DONE at 300.00) [time 300] Run job 2 for 300.00 secs (DONE at 600.00)

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19

Entry: 9

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF, Job 0 -- Response: 0.00 Turnaround 100.00 Job 1 -- Response: 100.00 Turnaround 300.00 Job 2 -- Response: 300.00

FIFO, Job 0 -- Response: 0.00 Turnaround 100.00 Job 1 -- Response: 100.00 Turnaround 300.00 Job 2 -- Response: 300.00

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100,

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

13.0

Score 2:

13.0

Score 3:

13.0

Entry: 10

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

With SJF: Job 0: Response \u2013 0, turnaround \u2013 100 Job 1: Response \u2013 100, turnaround \u2013 300 Job 2: Response \u2013 300, turnaround \u2013 600

With FIFO is the same because jobs arrive in the order of increasing range which makes the first come the shortest job.

However if we change the order of the jobs, the output will be different. For example with `-l 300,200,100` FIFO will be: Job 0: Response \u2013 300, turnaround \u2013 600 Job 1: Response \u2013 600, turnaround \u2013 900 Job 2: Response \u2013 900, turnaround \u2013 1200

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order.

However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19

Entry: 11

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

The average response time, turnaround time and waiting time may increase.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

3

Score 2:

3

Score 3:

3.0

Score outlier:

19

Entry: 12

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

```
python2 ./scheduler.py -p SJF -l 100,200,300 ARG policy SJF ARG jlist 100,200,300 Here is the job list, with the run time
python2 ./scheduler.py -p FIFO -l 100,200,300 ARG policy FIFO ARG jlist 100,200,300 Here is the job list, with the run time
```

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100,

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

13.0

Score 2:

13.0

Score 3:

13.0

Entry: 13

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF: Response Times: 0, 100, 300 Turnaround Times: 100, 300, 600 . FIFO: Response Times: 0, 100, 300 Turnaround Times: 100, 200, 600

If change order to be 200,100,300. SJF: Response Times: 100, 0, 300. Turnaround Times: 300, 100, 600.

If change order to 200,100,300 FIFO Response Times: 0, 200, 300 Turnaround Times: 200, 300, 600

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

- 1.For correct "Average response time and Turnaround time" for SJF (6.5 points)
- 2.For correct "Average response time and Turnaround time" for FIFO (6.5 points)
- 3.For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
- 4.For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19.0

Score 2:

19.0

Score 3:

19.0

Entry: 14

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For the initial order, FIFO and SJF have the same results, as the first jobs are the shortest. Response time = $0 + 100 + 300$

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19.0

Score 2:

19.0

Score 3:

19.0

Entry: 15

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

change the order to 200,100,300

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

0

Score 2:

0

Score 3:

0

Entry: 16

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF doesn't have any effect when changing the order of the length FIFO does cost more time in the order (300,200,100)

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

7.0

Score 2:

7.0

Score 3:

6.0

Entry: 17

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For SJF and FIFO, with the order 100, 200, 300, their result is same.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

1

Score 2:

0

Score 3:

0

Entry: 18

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For SJF Job 0 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 1 -- Response: 100.00 Turnaround 300.00 Wait 100.00

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19.0

Score 2:

19.0

Score 3:

19

Entry: 19

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

The result with SJF is still the same as with FIFO. If the order is changed, the result may be different. If the order is 200,100,300, the results for SJF policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results for SJF policy in different orders will change.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results for FIFO policy in different orders will change.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

6.5

Score 2:

6.5

Score 3:

6

Entry: 20

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

**** Solutions **** SJF will order the processes from the shortest to the longest, and this makes all the output the same no matter the order of the jobs.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19.0

Score 2:

19.0

Score 3:

19.0

Entry: 21

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

Same answers on SJF and FIFO with the original order of 100,200,300. It doubles the response and wait time and increases the turnaround time.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

6

Score 2:

6

Score 3:

6

Entry: 22

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

In principle, the job that arrives first will have the shortest response time and turnaround time while the last-in job will have the longest.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

7.0

Score 2:

7.0

Score 3:

8

Entry: 23

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For SJF, the order of jobs doesn't matter. It does not affect the response time, turnaround time and wait time, because the jobs are scheduled in order of their lengths.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

8

Score 2:

6

Score 3:

6

Entry: 24

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

When the order is 300,200,100, the average response, turnaround, and wait time of FIFO is 266.67, 466.67, and 266.67,

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100,

Sample criteria:

- 1.For correct "Average response time and Turnaround time" for SJF (6.5 points)
- 2.For correct "Average response time and Turnaround time" for FIFO (6.5 points)
- 3.For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
- 4.For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19

Entry: 25

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `(./scheduler.py -p SJF -l 100,200,300)`

Answer:

When the order of the job length is 100,200,300:

SJF: Job 1: response time=0, turnaround time=100

■Job 2: response time=100, turnaround time=300

■Job 3: response time=300, turnaround time=600

the average response time=133.33, the average turnaround time=333.33.

FIFO: Job 1: response time=0, turnaround time=100

■Job 2: response time=100, turnaround time=300

■Job 3: response time=300, turnaround time=600

the average response time=133.33, the average turnaround time=333.33.

Only in this order of length are response time and turnaround time the same for every job in SJF or FIFO policy. When the order is different, the results will change.

When the order of the job length is 200,100,300:

SJF: Job 1: response time=100, turnaround time=300

■Job 2: response time=0, turnaround time=100

■Job 3: response time=300, turnaround time=600

the average response time=133.33, the average turnaround time=333.33.

FIFO: Job 1: response time=0, turnaround time=200

■Job 2: response time=200, turnaround time=300

■Job 3: response time=300, turnaround time=600

the average response time= 166.67, the average turnaround time= 366.67.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1.For correct "Average response time and Turnaround time" for SJF (6.5 points)

2.For correct "Average reponse time and Turnaround time" for FIFO (6.5 points)

3.For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the

4.For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

13.0

Score 2:

13.0

Score 3:

13.0

Entry: 26

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF is the same result in question 1 FIFO is the same result in question 1

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

0

Score 2:

0

Score 3:

0

Entry: 27

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

Job 1 has a response time of 0, Turnaround time of 100, and a waiting time of 0. Job 2 has a response time of 100, Turnaround time of 200, and a waiting time of 100. Job 3 has a response time of 200, Turnaround time of 300, and a waiting time of 200.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19.0

Entry: 28

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For SJF, the value does not change as it always serve the shortest job in ascending order. But for FIFO, the largest the first job is, the larger the value.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

8

Score 2:

6

Score 3:

6

Score outlier:

19

Entry: 29

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

Response, Turnaround and Wait times are the same if the job list is 100,200,300 (this is because the jobs are sorted with their lengths)

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19

Entry: 30

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

If we use FIFO and try different order, the average Response and Turnaround time will change. The time will be shorter if we use SJF.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

7.0

Score 2:

7.0

Score 3:

6

Entry: 31

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

SJF: Job 0 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 1 -- Response: 100.00 Turnaround 300.00 Wait 100.00 Job 2 -- Response: 200.00 Turnaround 600.00 Wait 200.00

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

13.0

Score 2:

13.0

Score 3:

13.0

Entry: 32

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300` and `./scheduler.py -p FIFO -l 100,200,300`.

Answer:

If we do not change the order of the job lengths, both command will have the same result. For Job 0, the response time is 100, 200, 300. For Job 1, the response time is 200, 300, 400. For Job 2, the response time is 300, 400, 500. However, if we change the order of the job length to 300, 200, 100, the results of the two policies will be different. For SJF, the response time is 100, 200, 300. For FIFO, the response time is 300, 400, 500.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results for FIFO policy will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19

Entry: 33

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

`python2 scheduler.py -p SJF -l 100,200,300 -c Job 0 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 1 -- Response: 0.00 Turnaround 100.00 Wait 0.00 Job 2 -- Response: 0.00 Turnaround 100.00 Wait 0.00`

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19.0

Score 2:

19.0

Score 3:

19.0

Entry: 34

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For SJF, the order of the job length does not matter. For FIFO, the order of the job length would affect the average response time.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

6

Score 2:

6

Score 3:

6

Entry: 35

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

FIFO is always based on the order SJF if the 100 is at the second or third it will be execute first

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

0

Score 2:

0

Score 3:

0

Entry: 36

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For FIFO: The change of job length will not affect the calculation method of response time and turnaround time. The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same."
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

6.0

Score 2:

6.0

Score 3:

3.0

Entry: 37

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

In SJF, the job with a length of 100 will always be the first to do, followed by 200. The last job to do is 300. The order of the jobs is 100, 200, 300.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

6

Score 2:

6

Score 3:

6

Entry: 38

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

If the order of the job length is changed like 300,100,200, the response time and turnaround time will be: SJF: Job 0: 100, Job 1: 200, Job 2: 300

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the results will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

13.0

Entry: 39

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For SJF, The response time is 133.3 while the turnaround time is 333.33. When the order of job length is changed, the results for SJF policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the response time and turnaround time will be different.

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100, the response time and turnaround time will be different.

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same" (3 points)
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19

Entry: 40

Item: 1

Question:

Now do the same but with jobs of different lengths: 100, 200, and 300. The commands are `./scheduler.py -p SJF -l 100,200,300`

Answer:

For SJF the order of jobs doesn't matter because it anyway choosing the shortest job first. So response is 133.33 and turn

Sample answer:

The response time and turnaround time for three jobs using the FIFO policy and SJF policy are the same in the given order. However, the results for FIFO policy in different orders will change. In particular, if we run these jobs in order 300, 200, 100

Sample criteria:

1. For correct "Average response time and Turnaround time" for SJF (6.5 points)
2. For correct "Average response time and Turnaround time" for FIFO (6.5 points)
3. For point out "if change the order of job length, response time and turnaround time for the SJF policy should remain the same"
4. For correct explain "the results for FIFO policy in different orders will change." (3 points)

Full points:

19

Score 1:

19

Score 2:

19

Score 3:

19